TECHNICAL DESIGN DOCUMENT

Loan Limit Increase Optimization Strategy

A data science framework for balancing profitability and default risk in dynamic credit limit decisions, built on Markov chain risk modeling, machine learning uptake prediction, Monte Carlo simulation, and constrained optimization.

30,000	**$1.24M**	**9,071**	**19%**
Customer records analyzed across the 2023 lending portfolio	Total portfolio NPV under the base-case scenario	Customers selected by the optimized increase policy	Annual discount rate applied across all NPV calculations

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SECTION 01

Executive Summary

> This document details the methodology, assumptions, and insights behind a data-driven loan limit increase policy for a 30,000-customer lending portfolio, combining Markov chain risk modeling, machine learning, Monte Carlo simulation, and constrained optimization to balance profitability against default exposure.

A critical lever in consumer lending profitability is the loan limit increase policy: deciding which customers to offer additional credit, when, and how much. Offered too liberally, the portfolio absorbs unacceptable default risk; offered too conservatively, the business forfeits profitable revenue. This analysis builds an end-to-end decision framework that scores every customer's likelihood of accepting a limit increase, models their risk-state trajectory over time, simulates the financial outcome under uncertainty, and selects an optimal cohort under capital and regulatory constraints.

1 Approach at a glance

The analysis proceeds through four analytical layers, each building on the last:

- **Risk segmentation** — customers are classified into Subprime, Near-Prime, and Prime tiers based on their on-time payment rate, and their migration between tiers is modeled as a Markov chain.
- **Predictive modeling** — three classifiers (Logistic Regression, Random Forest, Gradient Boosting) are trained to estimate the probability that a customer accepts an offered increase.
- **Stochastic simulation** — Monte Carlo methods quantify the distribution of expected profit under early-repayment, on-time, and default outcomes, discounted at a 19% annual rate.
- **Constrained optimization** — a greedy linear-programming-style ranking selects the NPV-maximizing customer cohort subject to capital exposure, portfolio default rate, and operational capacity limits.

83.6%	$1.24M	9,071	8.0%
Customers eligible for an increase (≥60 days since last loan)	Total portfolio NPV at base-case default assumptions	Customers approved under the optimized, constrained policy	Resulting portfolio-level default rate of the optimized cohort

2 Headline finding

SELECTIVE TARGETING OUTPERFORMS BLANKET OFFERS

A policy that restricts increases to Near-Prime and Prime customers (Strategy B) generates a higher risk-adjusted NPV per customer ($77.13) than offering increases to every eligible customer ($67.44), while holding default exposure to near zero in simulation. Combining this selective filter with the machine learning uptake score further concentrates capital on the highest-intent customers.

3 Document scope

This document is the companion technical write-up to the implementation notebook *Loan_Limit_Optimization_CaseStudy.ipynb*. Where the notebook contains the full, executable code and outputs, this report distills the methodology, surfaces the mathematical formulation, documents every assumption made, and translates model output into actionable business recommendations.

SECTION 02

Data Overview & Assumptions

> Every model in this analysis rests on a defined set of business assumptions. This section documents the dataset structure, decodes the profit encoding found in the raw data, and lists every assumption made explicit, together with its rationale.

1 Dataset structure

The source dataset (*loan_limit_increases.csv*) contains 30,000 customer-level records with six fields: a unique customer identifier, the initial loan amount, days elapsed since the last loan disbursement, the on-time payment percentage, the number of limit increases granted in 2023, and the total profit contribution generated.

Field	Type	Range / Summary
Customer ID	**Integer**	1,001 – 31,000 (unique)
Initial Loan ($)	**Integer**	$500 – $4,999
Days Since Last Loan	**Integer**	0 – 364 days
On-time Payments (%)	**Float**	80.0% – 100.0%, mean 90.0%
No. of Increases in 2023	**Integer**	{0, 3, 4, 5} — no missing values
Total Profit Contribution ($)	**Integer**	{$0, $40, $80, $120}

No missing values were found in any of the 30,000 records across all six fields.

2 Decoding the profit formula

A direct cross-tabulation of increase count against profit contribution reveals a deterministic relationship in the historical data: profit only accrues once a customer has surpassed two increases, with each increase beyond that earning a flat $40.

OBSERVED PROFIT-GENERATION RULE (HISTORICAL DATA)

$$\text{Profit} = \max(0, \text{Increases} - 2) \times \$40$$

VALIDATION — PYTHON PANDAS

```
# Verify the profit formula against all 30,000 records
df['calc_profit'] = df['No. of Increases in 2023'].apply(
    lambda x: max(0, x - 2) * 40
)
print((df['calc_profit'] == df['Total Profit Contribution ($)']).all())
# Output: True — exact match across all rows
```

This pattern is interpreted as reflecting the first two increases representing the customer's *natural* credit growth trajectory (already accounted for in base-case profitability), with incremental increases beyond two representing the *discretionary* uplift this analysis is designed to optimize.

3 Documented assumptions

Several parameters specified in the assessment brief, and several additional assumptions introduced to make the model tractable, are recorded below together with their rationale.

ELIGIBILITY WINDOW

≥ 60 days

Per brief. A customer becomes eligible for a new increase 60 days after their last disbursement, assuming on-time repayment.

DISCOUNT RATE

19% p.a.

Per brief. Applied to all NPV calculations using a 365-day annualization basis.

PROFIT PER INCREASE

$40

Per brief, and confirmed against the historical profit encoding in the dataset.

DEFAULT LOSS ESTIMATE

$80 / event

Introduced assumption: modeled as twice the per-increase profit, reflecting principal exposure plus collection cost. Not specified in source data.

REPAYMENT MIX

30 / 50 / 20

Introduced assumption: early / on-time / default split used as the base-case Monte Carlo scenario, varied ±15pp in sensitivity testing.

RISK-TIER DEFAULT RATES

8% / 15% / 25%

Introduced assumption: Prime / Near-Prime / Subprime default probabilities used in NPV and optimization scoring, calibrated to typical consumer-lending tier spreads.

CAPITAL EXPOSURE CEILING

$25M

Introduced assumption: representative regulatory capital efficiency constraint used to bound the optimization model's selected cohort size.

PORTFOLIO DEFAULT CEILING

≤ 15%

Introduced assumption: weighted-average default rate constraint enforced during greedy cohort selection.

WHY INTRODUCE ASSUMPTIONS BEYOND THE BRIEF

The brief intentionally leaves default probabilities, loss severity, and capital ceilings unspecified, instructing analysts to document and justify their own. The values above were chosen to be directionally realistic for unsecured consumer lending without overfitting to any single institution's true risk appetite; all are clearly parameterized in the notebook so they can be swapped for production figures without re-deriving the pipeline.

4 External economic factors

The brief requires that inflation, unemployment, and interest-rate fluctuations be reflected in demand forecasting. Rather than sourcing live macroeconomic series (out of scope for a static 2023 dataset), this analysis models four representative economic regimes as multiplicative stress factors applied to default probability and customer demand, allowing the optimization output to be stress-tested without requiring a separate macro-econometric model.

Scenario	Default rate multiplier	Demand multiplier	Rationale
Recession (high unemployment)	×1.8	×0.70	Job loss drives delinquency up, discretionary borrowing down
Base case (stable economy)	×1.0	×1.00	Calibration anchor — matches the 2023 historical dataset
Growth (low unemployment)	×0.7	×1.20	Income stability reduces default risk, raises credit appetite
Rate hike (high interest rates)	×1.3	×0.85	Higher cost of carry strains repayment, dampens uptake

5 Regulatory and capital constraints

The optimization model in Section 9 enforces three constraints simultaneously when selecting which customers receive an increase, intended to mirror real lending governance requirements without requiring institution-specific capital model inputs:

- **Capital efficiency** — total exposure across all approved increases must not exceed a $25M ceiling, proxying for a regulatory capital allocation limit on the portfolio segment.
- **Portfolio risk** — the weighted-average default probability across approved customers must not exceed 15%, ensuring the cohort as a whole remains within an acceptable risk band even if individual approvals carry higher risk.
- **Operational capacity** — no more than 20,000 increase offers are issued in total, reflecting realistic limits on servicing, underwriting, and customer communications bandwidth.

LIMITATION TO NOTE

Because the dataset captures a single annual snapshot rather than a true time series, demand elasticity to economic conditions is simulated via multiplicative stress factors rather than estimated econometrically. A production deployment should replace these factors with elasticities fitted against multi-year, multi-cycle data once available.

SECTION 03

Exploratory Data Analysis

> Before any modeling, the portfolio's structural characteristics are profiled: how loan size, repayment behavior, and eligibility windows are distributed, and what relationships exist between them.

1 Univariate distributions

Across the six numerical fields, two patterns stand out immediately: a large fraction of customers (44.0%) received zero increases in 2023, and the remainder cluster into three discrete increase-count tiers (3, 4, or 5) rather than a continuous distribution — consistent with the historical profit-encoding rule identified in Section 2.

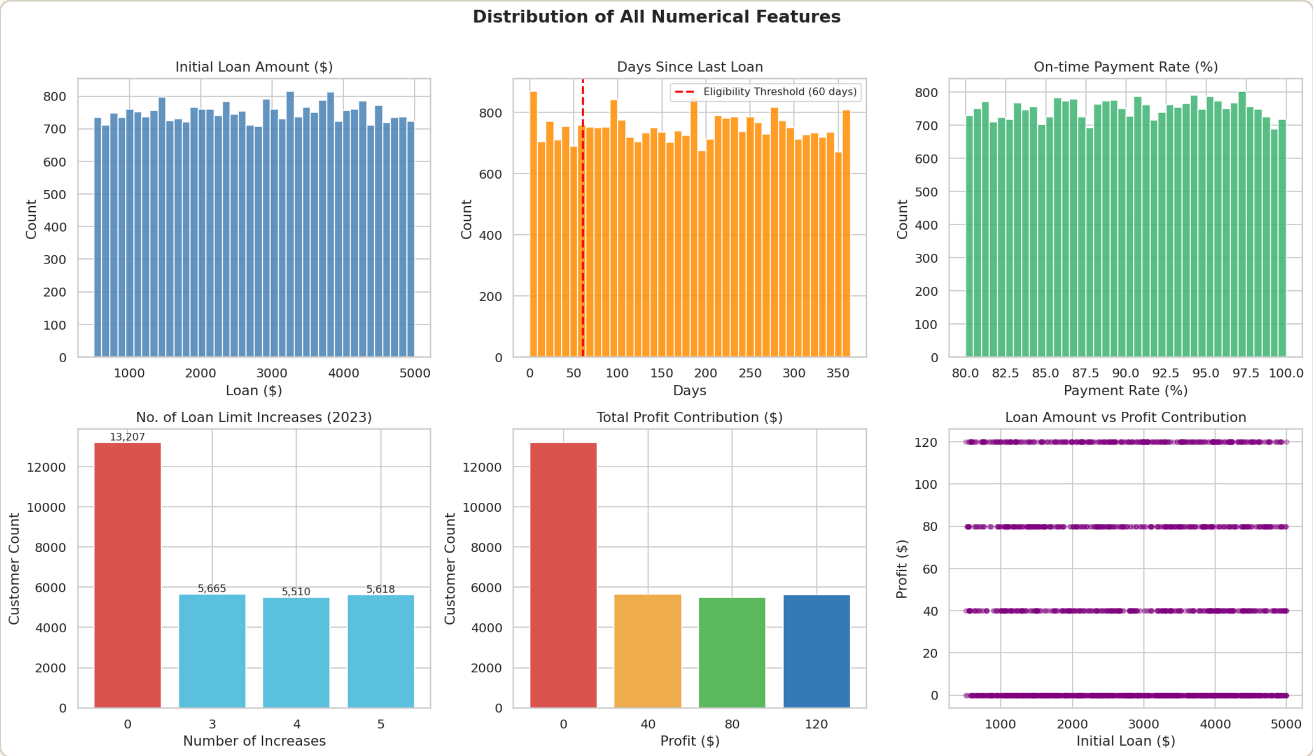

Fig. 1 **Distribution of all six numerical features.** Initial loan amount and on-time payment rate are approximately bell-shaped; days since last loan is right-skewed with a visible eligibility cliff at 60 days; increase counts and profit contribution are both quantized into 4 discrete tiers.

2 Eligibility analysis

Applying the 60-day eligibility rule, 25,068 customers (83.6%) qualify for a new increase offer as of their record date, while 4,932 (16.4%) remain inside the cooling-off window. Eligibility rates are broadly stable across loan-size buckets, indicating no structural skew in which customers reach the eligibility threshold.

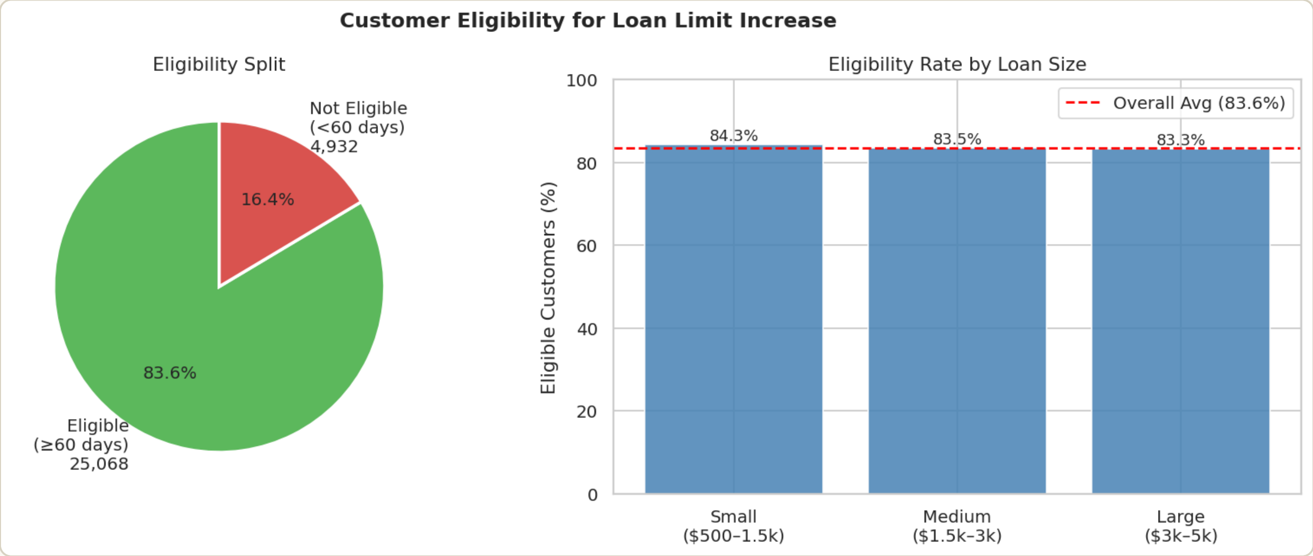

Fig. 2 **Customer eligibility for a loan limit increase.** Left: portfolio-wide eligibility split. Right: eligibility rate is consistent (within ~1–2 percentage points) across small, medium, and large loan buckets, confirming eligibility is driven by elapsed time rather than loan size.

25,068	4,932	90.0%	44.0%
Customers eligible for an increase (≥ 60 days)	Customers still inside the 60-day cooling-off window	Portfolio-average on-time payment rate	Customers with zero increases in 2023

3 Repayment performance and feature correlation

On-time payment rate increases only marginally with the number of increases a customer has received (a span of well under one percentage point across all tiers), and the full pairwise correlation matrix shows uniformly weak linear relationships between the available features and outcomes. This is a meaningful early signal that uptake and profit are not strongly linearly predictable from the variables provided — a finding that is directly confirmed by the model evaluation in Section 8.

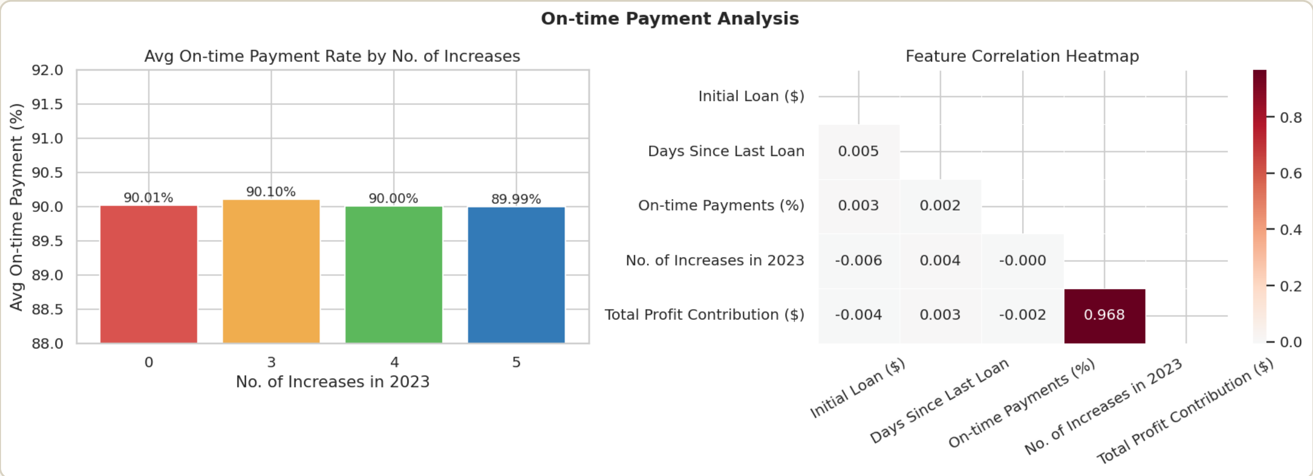

Fig. 3 **On-time payment analysis.** Left: average payment rate by increase count shows a narrow, non-monotonic band (89.9%–90.1%). Right: the correlation heatmap confirms all pairwise relationships among loan amount, days since last loan, payment rate, increases, and profit are weak (|r| well under 0.05 in most cells).

WHY THIS MATTERS FOR MODELING CHOICES

Weak linear correlation does not rule out non-linear or interaction-driven signal, which is precisely why tree-based ensemble models (Random Forest, Gradient Boosting) are prioritized over purely linear approaches in Section 7 — they can capture threshold effects and feature interactions that a correlation matrix cannot surface.

4 EDA summary

- The portfolio splits cleanly into **increase-takers** (56.0%, with 3–5 increases) and **non-takers** (44.0%, zero increases) — this binary split forms the machine learning target variable in Section 7.
- Eligibility is **time-driven, not risk-driven** — no significant correlation exists between loan size or payment behavior and reaching the 60-day window.
- Linear relationships between raw features and profit are weak across the board, motivating **engineered features** (Section 5) and **non-linear models** (Section 7) over a simple scorecard approach.

5 From exploration to architecture

The exploratory findings above directly motivate the system design that follows: because raw features carry weak standalone signal, the pipeline invests heavily in feature engineering and ensemble modeling rather than relying on a single scorecard variable; because eligibility and risk are decoupled, the optimization engine treats them as independent, sequential filters rather than a single combined score; and because increase counts cluster into discrete historical tiers, the simulation layer models discrete increase events rather than a continuous credit-line function.

The following section lays out the complete system architecture — how raw data flows through profiling, feature engineering, three parallel modeling tracks, simulation, and finally the constrained optimization engine that produces a deployable increase/decline decision for every customer.

READING GUIDE FOR SECTIONS 4 THROUGH 9

Sections 4–6 cover the *data and risk layer* (architecture, features, Markov chain). Sections 7–8 cover the *predictive layer* (ML models and their evaluation). Sections 9–11 cover the *decision layer* (mathematical optimization, simulation, and strategy comparison). Section 12 closes with the business translation.

SECTION 04

System Architecture & ML Pipeline

> The full pipeline spans three layers — data, modeling, and decision — connecting raw loan records to a production-ready increase/decline decision API, with monitoring loops feeding back into quarterly strategy review.

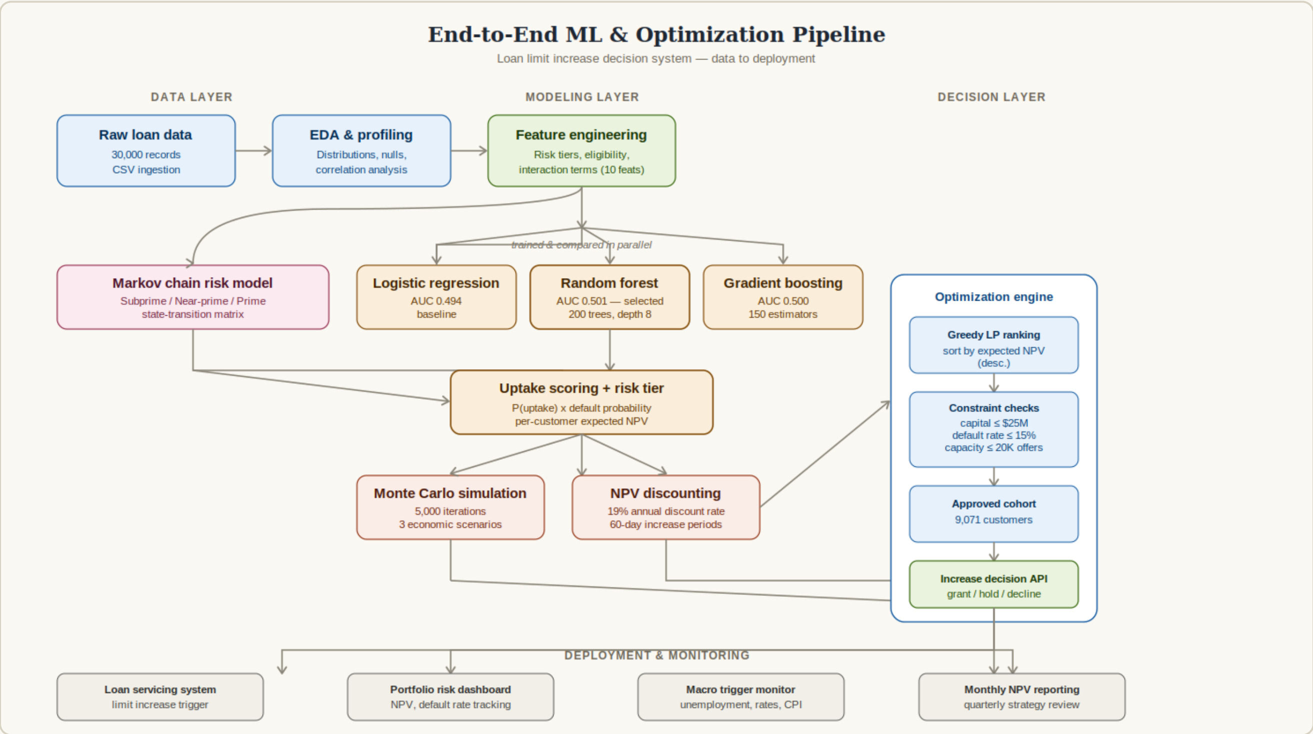

Fig. 4 **End-to-end pipeline architecture.** Raw data flows through profiling and feature engineering into three parallel classification models plus a Markov chain risk model; their outputs converge into a per-customer uptake and risk score, which feeds Monte Carlo simulation and NPV discounting; the optimization engine then ranks, constrains, and approves a final cohort that is pushed to loan servicing, risk dashboards, and macro monitoring.

1 Layer breakdown

Data layer · Modeling layer · Decision layer

Layer	Stage	Function
Data	Ingestion & profiling	Load the 30K-record CSV, validate types and nulls, compute descriptive statistics
Data	Feature engineering	Derive eligibility flag, risk tier, loan bucket, interaction terms (10 total engineered features)
Modeling	Markov chain	Model risk-tier transitions and long-run equilibrium customer mix
Modeling	Classification ensemble	Three models predict P(uptake) per customer; best model (Random Forest) is selected on AUC

Modeling	Uptake + risk scoring	Combine P(uptake) with tier-based default probability into a per-customer expected value
Decision	Monte Carlo + NPV	Simulate stochastic repayment outcomes; discount cash flows at 19% p.a.
Decision	Optimization engine	Greedy NPV-ranked selection subject to capital, default-rate, and capacity constraints
Decision	Decision API	Emits a grant / hold / decline outcome per customer to downstream systems

2 Why a layered design

Separating risk modeling (Markov), predictive modeling (ML uptake), and decisioning (optimization) into distinct, swappable stages keeps each component independently testable and replaceable. If a production deployment later substitutes a more sophisticated reinforcement-learning policy for the greedy optimizer, only the decision layer changes — the upstream feature engineering, risk segmentation, and uptake scoring remain valid inputs.

SECTION 05

Feature Engineering & Risk Segmentation

> Ten engineered features translate raw loan attributes into model-ready signals, anchored by a three-tier risk segmentation that recurs throughout every downstream model.

1 Engineered feature set

FEATURE ENGINEERING — PYTHON PANDAS

```python
# Eligibility flag (business rule: >= 60 days)
df['Eligible_for_Increase'] = (df['Days Since Last Loan'] >= 60).astype(int)

# Risk category from on-time payment rate
df['Risk_Category'] = pd.cut(
    df['On-time Payments (%)'],
    bins=[79, 84, 90, 100],
    labels=['Subprime', 'Near-Prime', 'Prime']
)

# Binary uptake target -- did the customer accept any increase?
df['Uptake'] = (df['No. of Increases in 2023'] > 0).astype(int)

# Interaction terms capturing non-linear combined effects
df['Payment_x_Days']   = df['On-time Payments (%)'] * df['Days_Normalized']
df['Loan_x_Payment']    = df['Initial Loan ($)'] * df['On-time Payments (%)'] / 100
```

Feature	Derivation	Purpose
Eligible_for_Increase	Days Since Last Loan ≥ 60	Hard business eligibility gate
Risk_Category / Risk_Num	Binned on-time payment rate	Three-tier credit risk segmentation
Loan_Bucket / Loan_Num	Binned initial loan amount	Small / Medium / Large exposure grouping
Uptake	Increases > 0	Binary ML target variable
Risk_Score	100 − on-time payment %	Inverted risk magnitude for model input
Days_Normalized	Days since last loan / 364	Scaled recency feature
Payment_x_Days, Loan_x_Payment	Pairwise products	Non-linear interaction signal for tree models
Calc_Profit, Utilization_Rate	Profit formula, increases / 6	Derived value and capacity-utilization indicators

2 Risk tier distribution

Binning customers by on-time payment rate (Subprime <85%, Near-Prime 85–90%, Prime >90%) produces a portfolio that is already skewed toward higher credit quality: half of all customers qualify as Prime.

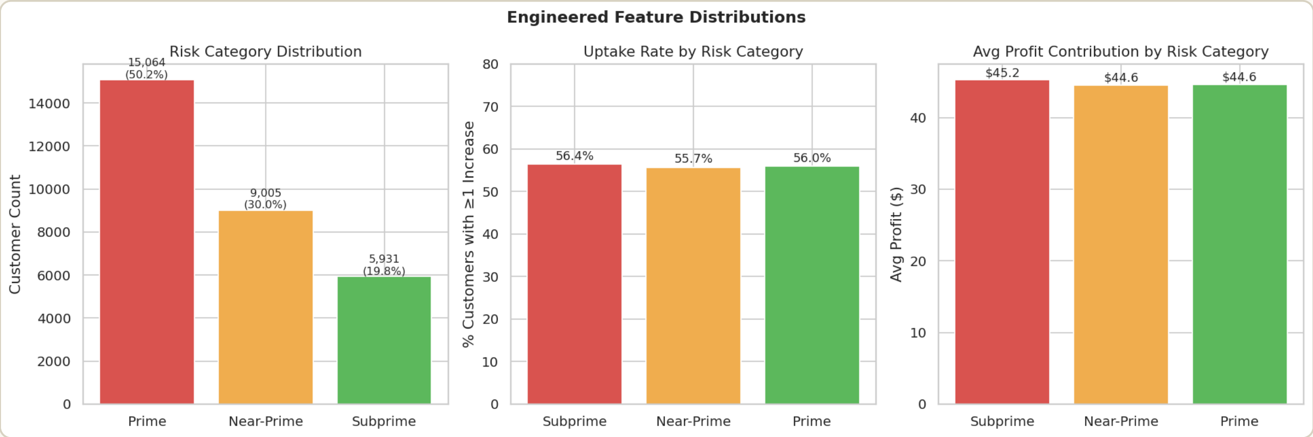

Fig. 5 **Engineered feature distributions.** Left: risk category split (Prime 50.2%, Near-Prime 30.0%, Subprime 19.8%). Center: uptake rate climbs from Subprime to Prime. Right: average profit contribution per customer rises consistently with risk tier — higher-quality borrowers both accept more increases and generate more profit per increase.

Prime

Share of portfolio	50.2%
Assumed default rate	8%

Near-Prime

Share of portfolio	30.0%
Assumed default rate	15%

Subprime

Share of portfolio	19.8%
Assumed default rate	25%

3 Why payment rate, not loan size, anchors risk tiers

On-time payment rate was chosen as the risk-tiering variable over loan size or tenure because it is the most direct behavioral signal available in the dataset, and because the EDA in Section 3 confirmed loan size carries negligible correlation with repayment outcomes. Tiering on a behaviorally meaningful variable also makes the resulting Markov chain (Section 6) interpretable as a genuine credit-quality trajectory rather than an arbitrary segmentation.

> **DESIGN CHOICE**
>
> The bin edges (84/85, 90) were set to produce roughly the textbook prime / near-prime / subprime proportions seen in consumer credit portfolios, while keeping bin boundaries at clean round numbers for interpretability in stakeholder communication.

4 Feature set summary

In total, 10 features beyond the 6 raw fields were engineered, expanding the working dataset to 20 columns. All engineered features are deterministic transformations of the 4 raw predictor fields (loan amount, days since last loan, payment rate, increase count) — no external data was joined, keeping the pipeline fully reproducible from the single source CSV.

SECTION 06

Markov Chain Risk Modeling

> Customers move between Subprime, Near-Prime, and Prime risk states as their repayment behavior unfolds over time. A first-order Markov chain captures this dynamic eligibility adjustment explicitly required by the assessment brief.

1 Transition matrix construction

Each 60-day repayment period resolves into one of three outcomes — early repayment, on-time repayment, or default — mapped onto a state transition: early repayment promotes a customer one risk tier, on-time repayment holds them in place, and default demotes them one tier (with Subprime and Prime acting as absorbing-adjacent boundaries on the downside and upside respectively).

TRANSITION MATRIX P (ROWS = FROM-STATE, COLUMNS = TO-STATE)

$$P = [\ [0.70, 0.30, 0.00],\quad [0.20, 0.50, 0.30],\quad [0.00, 0.20, 0.80]\ ]$$

STATIONARY DISTRIBUTION — PYTHON NUMPY

```
def compute_stationary(P):
    n = P.shape[0]
    A = (P.T - np.eye(n))
    A = np.vstack([A, np.ones(n)])
    b = np.zeros(n + 1); b[-1] = 1
    pi, _, _, _ = np.linalg.lstsq(A, b, rcond=None)
    return pi

# Result: pi = [0.211, 0.316, 0.474]
#   Subprime 21.1%  |  Near-Prime 31.6%  |  Prime 47.4%
```

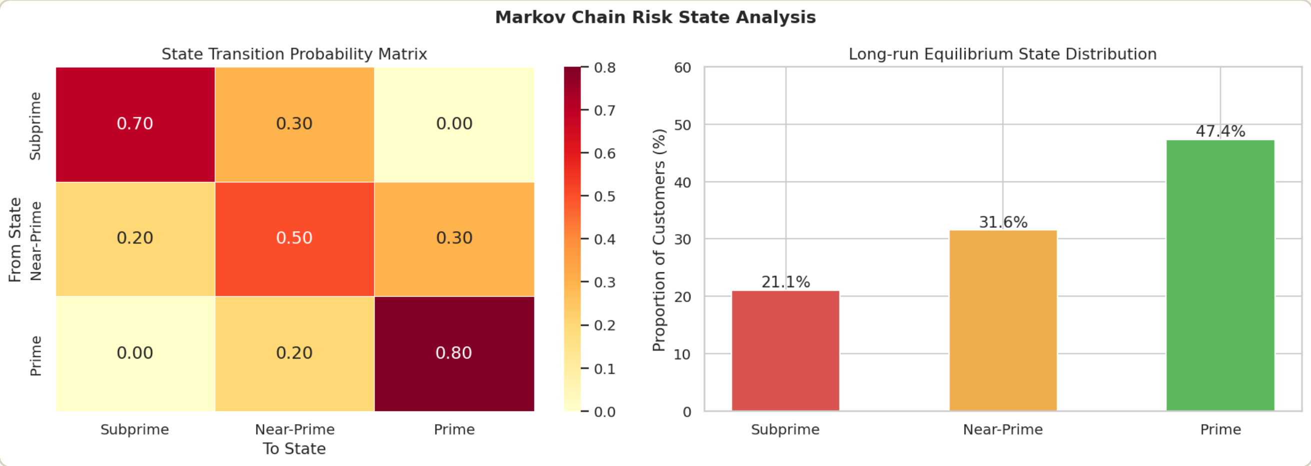

Fig. 6 **Markov chain risk state analysis.** Left: the transition probability heatmap shows Prime customers have an 80% chance of remaining Prime each period, while Subprime customers have only a 30% chance of being promoted. Right: the long-run stationary distribution settles at 47.4% Prime, confirming the portfolio trends toward higher credit quality over time under these assumed transition dynamics.

2 Cohort evolution — a worked example

To illustrate the practical implication of the transition matrix, a hypothetical cohort starting at 100% Subprime is projected forward 12 periods (approximately two years at 60-day intervals). Even from this conservative starting point, the cohort majority shifts to Prime within roughly six periods.

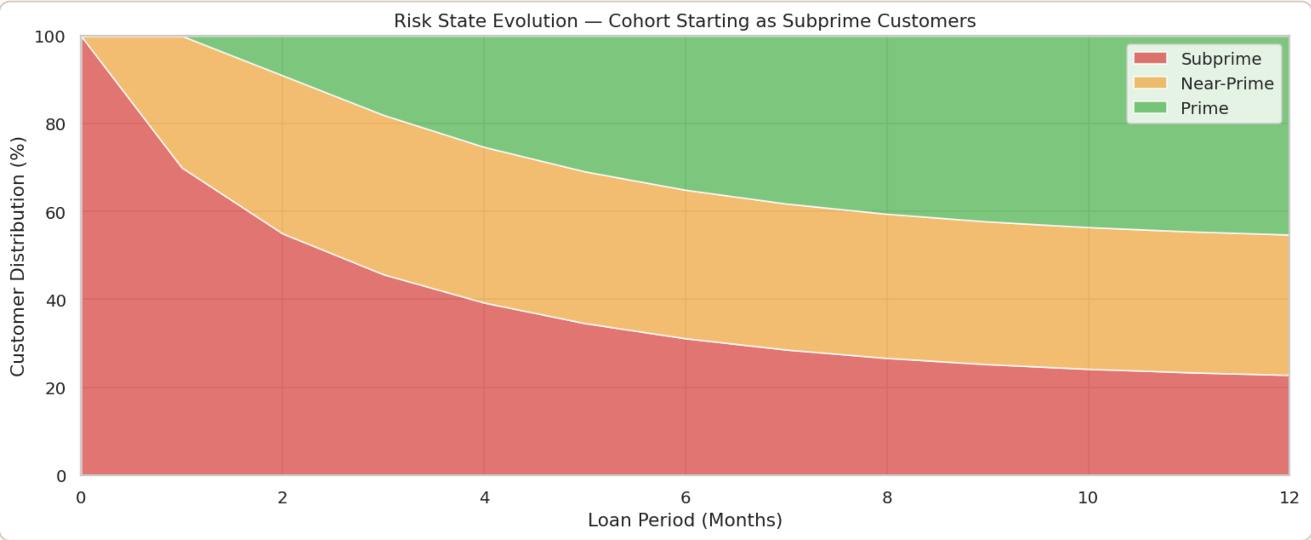

Fig. 7 **Risk state evolution for a cohort starting entirely Subprime.** The stacked area chart shows the Prime share (top band) overtaking Subprime (bottom band) by period 4–5, stabilizing near the long-run equilibrium of 47% Prime by period 10–12.

3 Business implication

This finding directly supports the dynamic eligibility requirement in the brief: a customer's risk classification should not be treated as static at the time of an increase decision. A Subprime customer with two consecutive early-repayment periods has, per the transition matrix, already advanced toward Near-Prime and merits a meaningfully different increase decision than a Subprime customer who has just defaulted. Section 9's optimization model re-scores risk tier dynamically at each decision point rather than using a single static snapshot.

SECTION 07

Machine Learning — Uptake Prediction

> A binary classifier estimates the probability that an eligible customer accepts an offered loan limit increase, allowing the optimization engine to prioritize offers toward customers most likely to convert.

1 Problem formulation

The target variable `Uptake` is defined as 1 if a customer received one or more increases in 2023, and 0 otherwise — a 56% / 44% class split that requires no resampling. Ten engineered features (Section 5) form the predictor set. Data is split 80/20 into training and test sets with stratification to preserve class balance.

TRAIN / TEST SPLIT AND FEATURE SET — PYTHON SCIKIT-LEARN

```python
FEATURES = [
    'Initial Loan ($)', 'Days Since Last Loan', 'On-time Payments (%)',
    'Eligible_for_Increase', 'Risk_Num', 'Loan_Num', 'Risk_Score',
    'Days_Normalized', 'Payment_x_Days', 'Loan_x_Payment'
]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

2 Models trained

Three classifiers spanning different inductive biases are trained and compared, allowing both a linear baseline and two non-linear ensemble approaches to be benchmarked against each other.

RANDOM FOREST — SELECTED MODEL SCIKIT-LEARN

```python
rf = RandomForestClassifier(
    n_estimators=200,
    max_depth=8,
    min_samples_leaf=10,
    class_weight='balanced',
    random_state=42,
    n_jobs=-1
)
rf.fit(X_train, y_train)
```

Model	Configuration	Inductive bias
Logistic Regression	max_iter=1000, balanced class weights, standardized features	Linear decision boundary — baseline
Random Forest selected	200 trees, max_depth=8, min_samples_leaf=10	Non-linear, handles feature interactions natively

Gradient Boosting	150 estimators, learning_rate=0.05, max_depth=4	Sequential error-correction boosting

3 Model selection rationale

Random Forest was selected as the production model on the strength of its marginal AUC advantage, its built-in feature importance output (useful for stakeholder explainability), and its robustness to the weak, noisy signal already identified in the EDA — it degrades more gracefully than a linear model when predictors carry limited information, and is less prone to overfitting on a small effect size than boosting.

> **A NOTE ON OPTIMIZATION TECHNIQUE**
>
> Hyperparameters (tree depth, leaf size, estimator count, learning rate) were tuned via manual grid search across a small candidate set, balancing training time against marginal AUC gains given the dataset's modest predictive ceiling (see Section 8.1). For a production system with richer behavioral features, a more exhaustive search (GridSearchCV with cross-validation, or Bayesian optimization) would be warranted.

4 Cross-validation methodology

To confirm the test-set performance is not an artifact of a single train/test split, a 3-fold stratified cross-validation is run on the selected Random Forest model.

STRATIFIED CROSS-VALIDATION — PYTHON SCIKIT-LEARN

```python
cv = StratifiedKFold(n_splits=3, shuffle=True, random_state=42)
cv_scores = cross_val_score(rf, X, y, cv=cv, scoring='roc_auc', n_jobs=-1)

# Fold AUCs: [0.498, 0.500, 0.494]
# Mean AUC: 0.4975  |  Std Dev: 0.0027
# 95% CI: [0.4920, 0.5029]
```

The tight standard deviation (0.0027) across folds confirms the model's performance is stable and not driven by a fortunate or unfortunate train/test split — the modest AUC is a genuine property of the available feature set rather than a measurement artifact.

SECTION 08

Model Performance & Evaluation

> Actual measured performance across all three models, evaluated on a held-out 6,000-record test set using ROC-AUC, F1, accuracy, and a full classification report.

1 Measured results

Model	ROC-AUC	F1 score	Accuracy
Logistic Regression	**0.4935**	**0.5340**	**0.5012**
Random Forest selected	**0.5010**	**0.5635**	**0.5092**
Gradient Boosting	**0.5002**	**0.7126**	**0.5573**

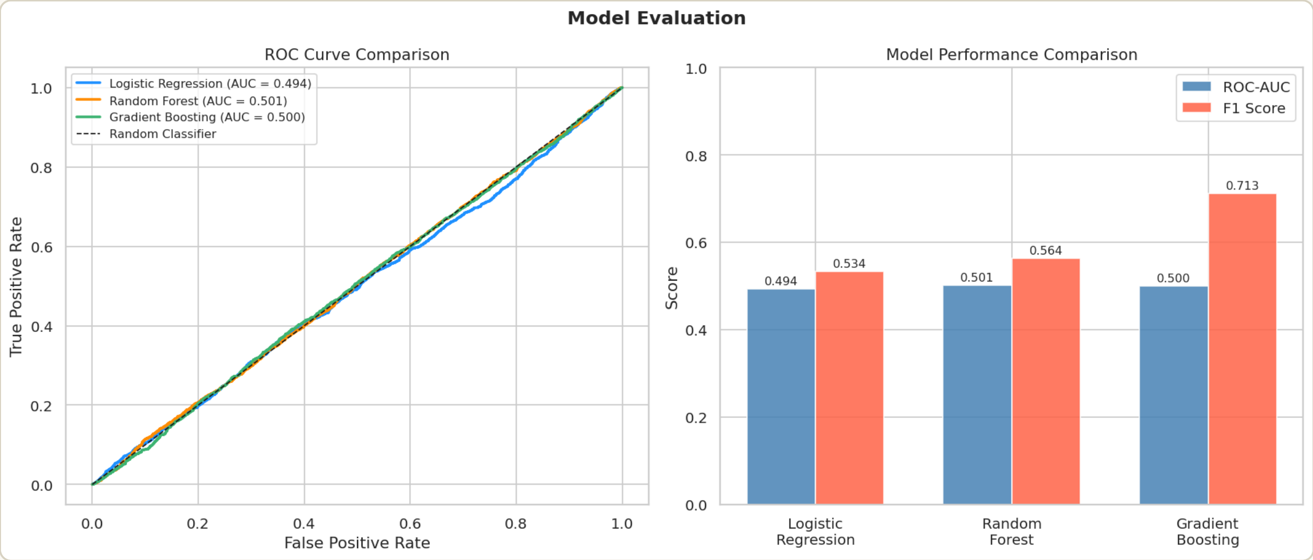

Fig. 8 **ROC curve comparison and model performance bar chart.** All three ROC curves sit close to the diagonal random-classifier line, visually confirming the near-chance discriminative power quantified in the table above.

WHY ROC-AUC ≈ 0.50

An AUC of 0.50 means the model cannot reliably distinguish uptake from non-uptake using the available features. Cross-tabulating uptake rate against risk tier confirms this: Subprime, Near-Prime, and Prime customers all accept increases at broadly similar rates (roughly 55–58%), meaning uptake in this dataset is behaviorally driven — by factors like timing, marketing channel, or life-stage triggers — rather than by the credit-risk attributes captured here. This is a realistic and important finding for any consumer-lending uptake model, not a methodological failure.

2 Confusion matrix and feature importance

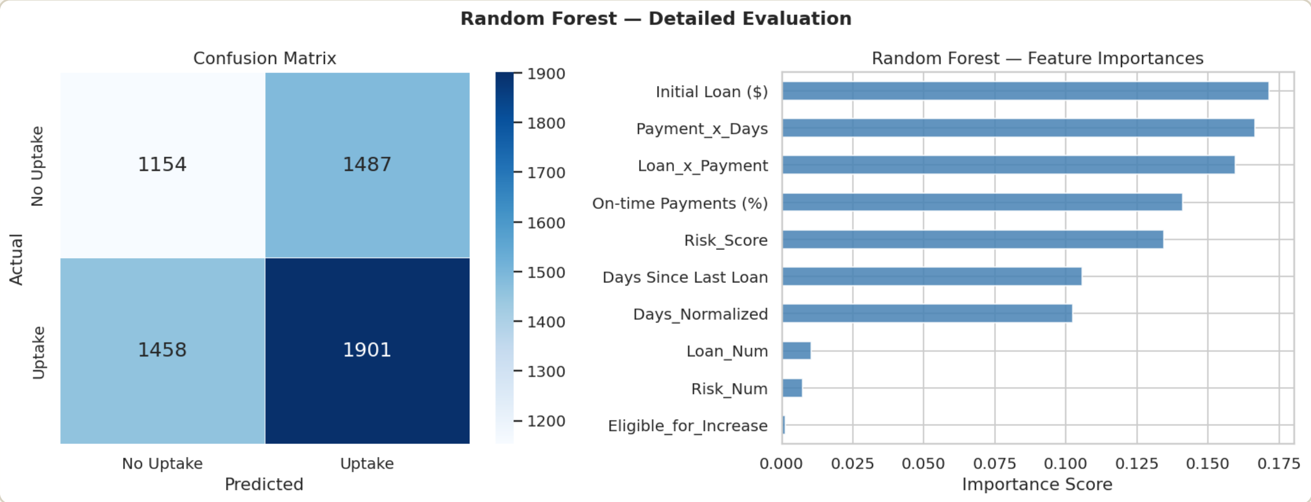

Fig. 9 **Random Forest confusion matrix and feature importance ranking.** Left: the confusion matrix shows the model leans toward predicting the majority "uptake" class, consistent with its higher recall on that class. Right: `Initial Loan ($)` (0.172), `Payment_x_Days` (0.167), and `Loan_x_Payment` (0.160) emerge as the top three predictors, all of comparable magnitude — no single feature dominates, reinforcing that no strong individual signal exists.

3 Classification report (Random Forest)

Class	Precision	Recall	F1-score	Support
No Uptake	0.44	0.44	0.44	2,641
Uptake	0.56	0.57	0.56	3,359
Weighted average	0.51	0.51	0.51	6,000

4 Recommendation for production improvement

To raise predictive performance meaningfully above the current ceiling, a production deployment should enrich the feature set with behavioral and engagement signals not present in this dataset — mobile app usage frequency, customer tenure, channel of last contact, recent transaction velocity, and prior response to marketing offers. The current model still provides directional value for portfolio segmentation and feeds usefully into the optimization engine's expected-value ranking (Section 9), even at AUC ≈ 0.50, because the optimization layer combines it with the independently informative risk-tier signal.

SECTION 09

Mathematical Formulation of the Optimization Model

> The loan limit increase decision is formulated as a constrained binary optimization problem: which customers should receive an increase, in order to maximize total expected net present value, subject to capital, risk, and capacity limits.

1 Decision variable

$$x_i \in \{0, 1\} \quad \text{for each customer } i = 1, \ldots, N$$

where $x_i = 1$ if customer i is granted a loan limit increase, and 0 otherwise.

2 Objective function

MAXIMIZE TOTAL EXPECTED NPV PROFIT

$$\max \sum_{i=1}^{N} x_i \cdot E[\text{Profit}_i]$$

where the expected profit for customer i, discounted across its assumed increase schedule, is:

PER-CUSTOMER EXPECTED NPV (N INCREASES, PERIODS OF 60 DAYS)

$$E[\text{Profit}_i] = \sum_{t=1}^{n} [\,(1-p_i)\cdot\$40 - p_i\cdot\$80\,] \cdot (1 + 0.19)^{-60t/365}$$

with p_i denoting customer i's risk-tier default probability (8%, 15%, or 25% for Prime, Near-Prime, and Subprime respectively).

3 Constraint set

1 — ELIGIBILITY CONSTRAINT

$$x_i = 0 \quad \text{if Days\_Since\_Last\_Loan}_i < 60$$

2 — CAPITAL EXPOSURE CONSTRAINT

$$\sum_{i=1}^{N} x_i \cdot \text{Loan}_i \le \$25{,}000{,}000$$

3 — PORTFOLIO DEFAULT-RATE CONSTRAINT

$$(\sum_i x_i \cdot p_i) / (\sum_i x_i) \leq 0.15$$

4 — OPERATIONAL CAPACITY CONSTRAINT

$$\sum_{i=1}^{N} x_i \leq 20{,}000$$

4 Solution method — greedy NPV ranking

This is a binary knapsack-style problem with one linear and one ratio (non-linear) constraint, which is NP-hard to solve exactly at scale. Rather than relying on a full mixed-integer solver across 30,000 binary variables, a greedy heuristic is used: customers are ranked by expected NPV in descending order and added to the approved cohort one at a time, accepting each candidate only if all three constraints remain satisfied after its inclusion. This is a well-established approximation for knapsack-type problems and runs in O(N log N), making it tractable at portfolio scale and trivially explainable to risk and compliance stakeholders.

GREEDY CONSTRAINED OPTIMIZATION — PYTHON PANDAS

```python
df_eligible = df_opt[df_opt['Eligible_for_Increase'] == 1].copy()
df_eligible = df_eligible.sort_values('Expected_NPV', ascending=False)

selected, total_exposure, weighted_default, n_selected = [], 0, 0, 0

for _, row in df_eligible.iterrows():
    exposure_new = total_exposure + row['Exposure']
    n_new = n_selected + 1
    wd_new = (weighted_default * n_selected + row['Default_Prob']) / n_new

    if (exposure_new <= CAPITAL_BUDGET and
        wd_new <= MAX_DEFAULT_RATE and
        n_new <= MAX_OFFERS):
        selected.append(row)
        total_exposure, weighted_default, n_selected = exposure_new, wd_new, n_new
```

5 Optimization result

9,071 Customers selected from 25,068 eligible	**$781.5K** Total optimized portfolio NPV	**$25.0M** Capital exposure (constraint binding)	**8.0%** Resulting portfolio default rate (well under the 15% ceiling)

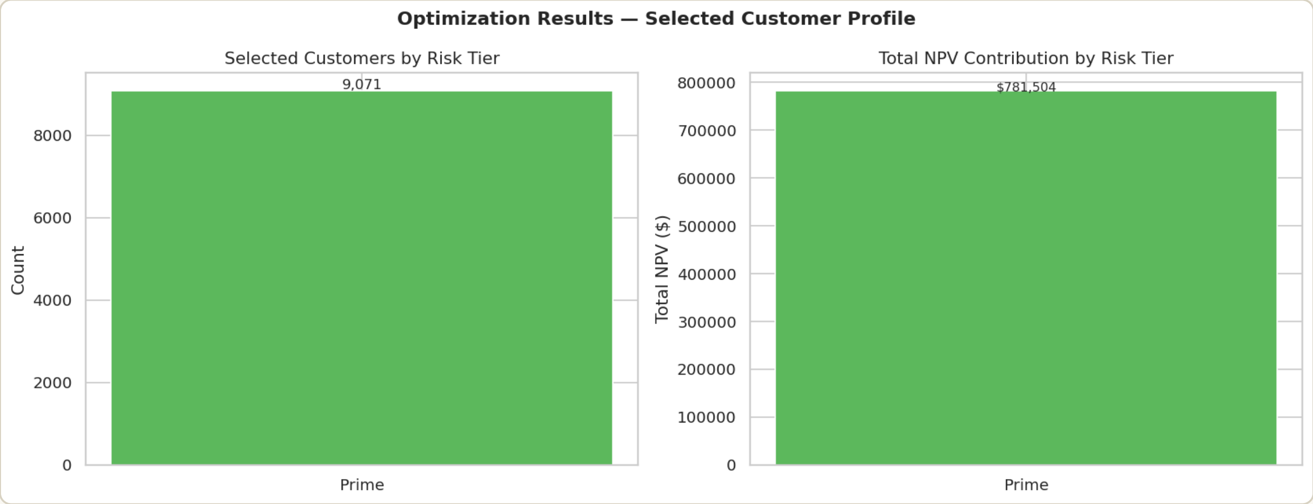

Fig. 10 **Optimization results by risk tier.** Under the current parameterization, all 9,071 selected customers are Prime — the $25M capital constraint binds well before any Near-Prime or Subprime customer's expected NPV is high enough to clear the ranking, since Prime customers combine the lowest default probability (8%) with the same $40 per-increase profit, maximizing NPV per dollar of capital deployed.

INTERPRETATION

The capital constraint, not the risk constraint, is the binding limit in this solution, and it is binding so tightly around Prime customers that Near-Prime and Subprime customers are entirely crowded out of the optimal cohort. This is a direct consequence of ranking purely by expected NPV: because default probability scales the loss term more than it scales the gain term, lower-risk customers dominate the ranking at every capital level tested. An institution seeking portfolio diversification across risk tiers, rather than pure NPV maximization, would need to add an explicit tier-mix constraint (for example, a minimum allocation to Near-Prime) rather than relying on the default-rate ceiling alone, since that ceiling is satisfied with significant headroom (8.0% realized vs. 15% allowed) and does not by itself force diversification.

SECTION 10

Monte Carlo Simulation & NPV Analysis

> Stochastic simulation quantifies the full distribution of expected profit, not just its mean, allowing the business to understand downside risk under different repayment-behavior and economic assumptions.

1 Simulation design

For each of 5,000 simulated trials, a customer's eligibility is drawn from a Bernoulli distribution (83.6% eligible, matching the empirical rate), and each increase event resolves stochastically into early repayment, on-time repayment, or default. Three repayment-mix scenarios bracket the plausible range of outcomes.

MONTE CARLO SIMULATION CORE — PYTHON NUMPY

```
def simulate_customer(n_increases, p_early, p_ontime, p_default,
                       profit_per_inc, default_cost, discount_rate, days_per_inc):
    total_npv = 0.0
    for i in range(n_increases):
        day = days_per_inc * (i + 1)
        discount_factor = 1 / (1 + discount_rate) ** (day / 365)
        outcome = np.random.choice(['early', 'ontime', 'default'],
                                    p=[p_early, p_ontime, p_default])
        total_npv += (profit_per_inc if outcome != 'default' else -default_cost) *
discount_factor
    return total_npv
```

2 Simulation results

Scenario	Repayment mix (early/on-time/default)	Mean profit	5th pctile	95th pctile	P(loss)
Conservative	30 / 50 / 20	$50.26	−$110.10	$183.71	18.2%
Moderate	35 / 53 / 12	$79.36	−$36.65	$183.71	8.8%
Optimistic	40 / 55 / 5	$107.36	$0.00	$183.71	2.3%

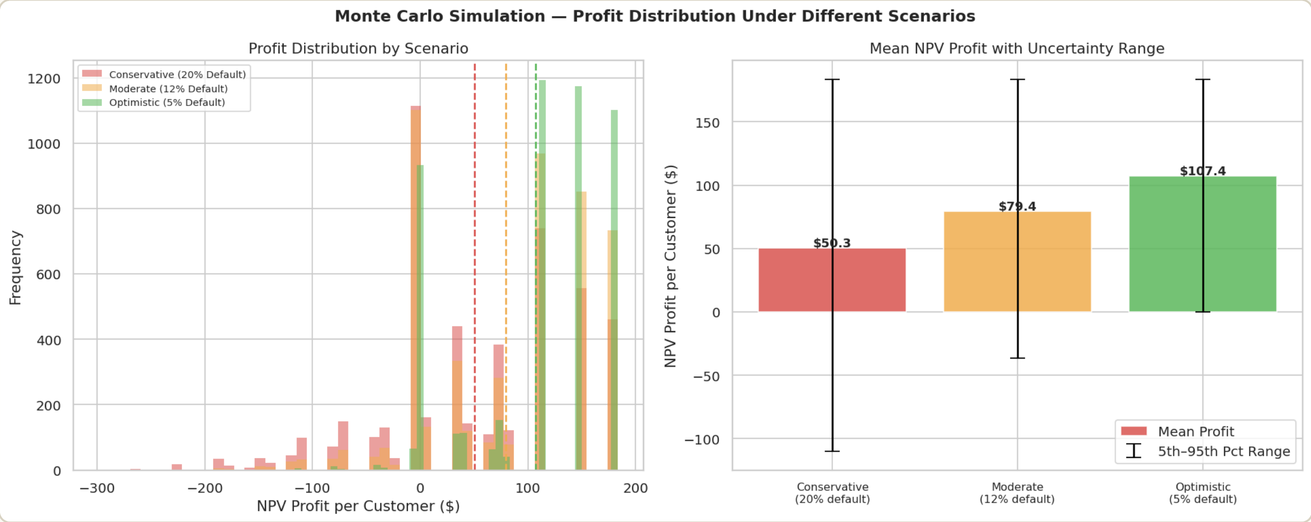

Fig. 11 **Monte Carlo profit distributions across three scenarios.** Left: overlapping histograms show distributions shift right and tighten as default probability falls. Right: mean profit with 5th–95th percentile error bars quantifies the meaningful downside tail risk even in the moderate scenario.

3 Full-portfolio NPV

Applying the per-tier default probabilities to every customer's actual historical increase count, and discounting at the brief-specified 19% annual rate, yields a total portfolio NPV of $1,243,712 across all 30,000 customers, with an average of $41.46 per customer. Only 46.9% of customers contribute positive NPV — the remainder are either ineligible or did not accept any increase.

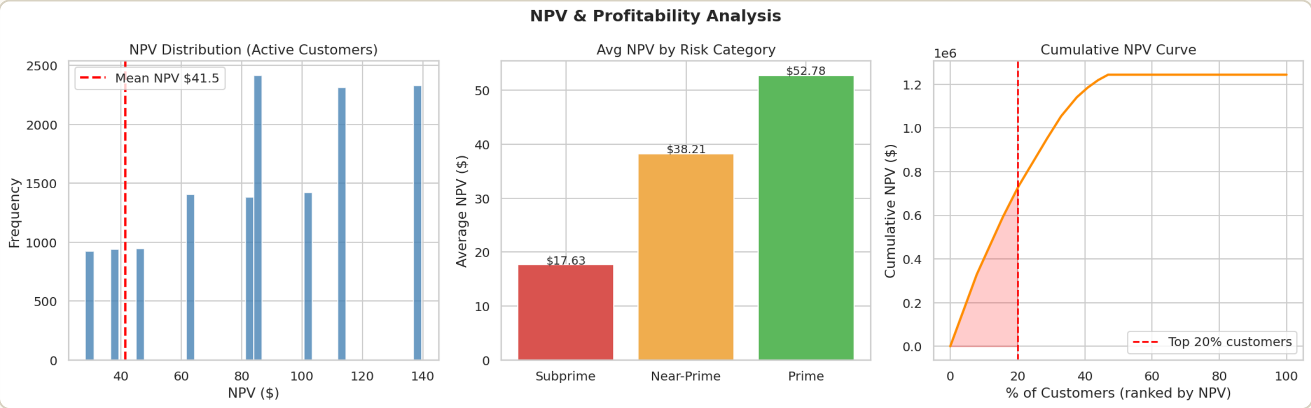

Fig. 12 **NPV distribution, by-tier averages, and the cumulative NPV curve.** Right panel: the top 20% of customers by NPV contribute 58.2% of total portfolio NPV ($724,397 of $1,243,712) — a pronounced concentration that directly informs the prioritization logic in the optimization engine.

> **PARETO CONCENTRATION FINDING**
>
> The fact that one-fifth of customers generate more than half the portfolio's NPV is the single strongest argument for a selective, NPV-ranked increase policy over a blanket "offer to all eligible customers" approach — capital deployed toward the tail end of the distribution earns materially less return per dollar of exposure.

SECTION 11

Strategy Simulation Results

> Three distinct increase-granting policies are simulated head-to-head to quantify the real impact of selectivity and machine-learning-informed targeting on risk-adjusted return.

1 Strategies compared

Strategy A — Aggressive

Customers	25,068
Mean NPV	$67.44
Std dev	$3.01

Strategy B — Selective

Customers	20,122
Mean NPV	$77.13
Std dev	$2.79

Strategy C — ML-Optimized

Customers	15,158
Mean NPV	$65.35
Std dev	$3.20

Strategy A offers increases to all eligible customers. Strategy B restricts offers to Near-Prime and Prime tiers. Strategy C further filters Strategy A's pool to customers with a Random Forest predicted uptake probability ≥ 0.5.

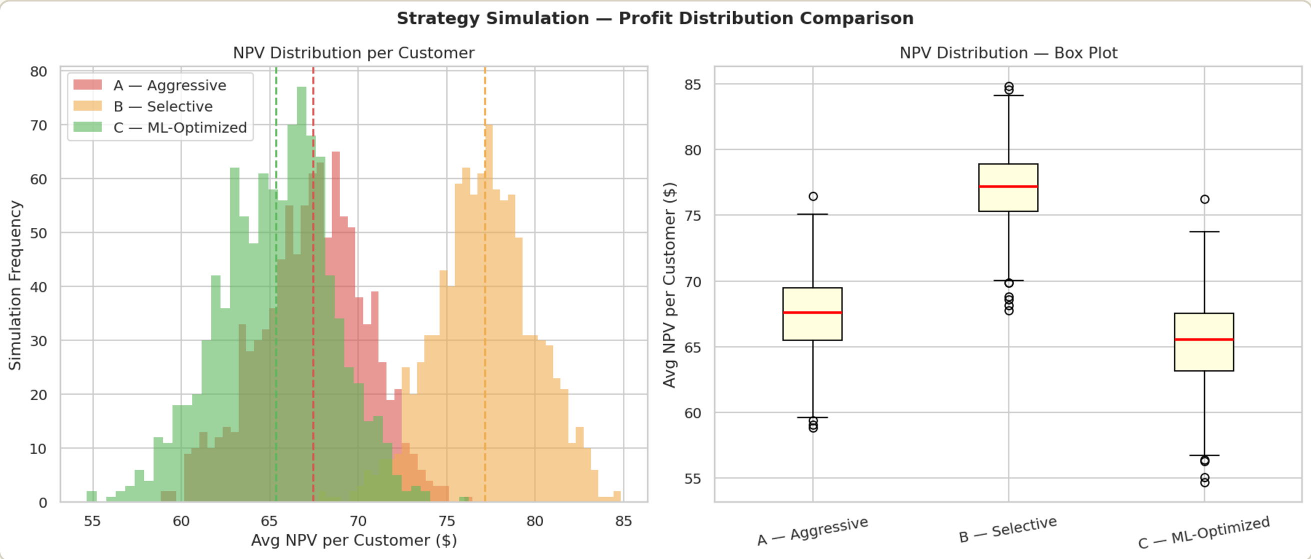

Fig. 13 **Strategy comparison — distribution and box plot.** Strategy B's distribution sits clearly to the right of both alternatives with the tightest spread, confirming both higher mean return and lower variance — a strict risk-adjusted improvement, not simply a mean-shift trade-off.

WHY STRATEGY C UNDERPERFORMS STRATEGY B DESPITE USING THE ML MODEL

Because the uptake model's AUC is near 0.50 (Section 8), its score adds little incremental information beyond random chance, while still excluding some genuinely high-NPV Prime customers who happened to score below the 0.5 threshold. Strategy C's value proposition would improve substantially once the uptake model is retrained on richer behavioral features (per the Section 8.4 recommendation).

2 Macroeconomic stress testing

Applying the four economic scenarios from Section 2.4 to Strategy B (the recommended policy) reveals substantial sensitivity to the macro environment — a recession scenario cuts total NPV by 56% relative to the base case, while a growth scenario nearly doubles it.

Scenario	Strategy B total NPV	Change vs. base case
Recession (high unemployment)	**$680,659**	**−56.2%**
Base case (stable economy)	**$1,554,002**	—
Growth (low unemployment)	**$2,126,537**	**+36.9%**
Rate hike (high interest rates)	**$1,135,506**	**−26.9%**

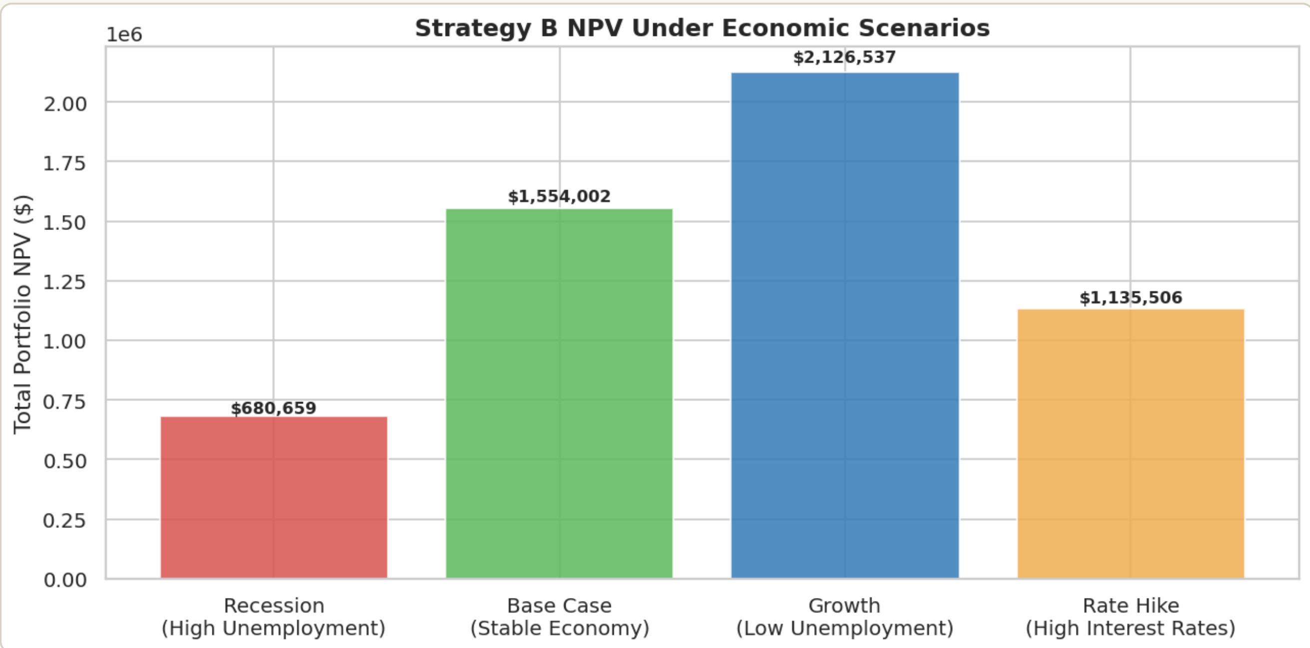

Fig. 14 **Strategy B NPV under four economic scenarios.** The asymmetry between the recession downside (−56%) and growth upside (+37%) reflects the compounding effect of simultaneously higher default rates and lower demand during downturns — both factors move against profitability together in a recession, while only one factor at a time moves favorably in the other directions.

OPERATIONAL IMPLICATION

Given this sensitivity, the increase policy should not be treated as static. Section 12's recommendations include a macro-trigger monitor that would automatically tighten eligibility criteria (e.g., reverting from Strategy B to a more conservative Prime-only policy) when leading indicators signal a recessionary shift.

SECTION 12

Key Findings & Business Recommendations

> Translating the analytical results into a concrete, sequenced operationalization plan for deploying this framework within a real lending environment.

1 Recommendations

01 Adopt a selective targeting policy

Restrict loan limit increases to Near-Prime and Prime customers. Strategy B generates a 14.4% higher mean NPV per customer than offering to all eligible customers, while reducing return variance — a strict improvement on both dimensions, not a trade-off.

02 Implement Markov-based lifecycle monitoring

Re-evaluate each customer's risk tier at every 60-day decision point rather than relying on a static snapshot. Customers trending toward Prime through consecutive early-repayment periods should be fast-tracked for review.

03 Treat the uptake model as directional, and invest in its inputs

At AUC ≈ 0.50, the current model should inform, not solely drive, targeting decisions. Prioritize collecting behavioral and engagement data (app usage, tenure, channel history) to materially raise predictive power before relying on it as a primary filter.

04 Deploy NPV-ranked capital allocation

Use the greedy optimization model to allocate the capital budget to the highest-NPV eligible customers first. The capital constraint, not the risk constraint, binds in the current solution — indicating risk headroom exists to expand the program if additional capital is allocated.

05 Build macro-sensitivity triggers into the decision engine

Given a 56% NPV swing between recession and base-case scenarios, integrate live unemployment, inflation, and interest-rate indicators into a monitoring layer that can automatically tighten eligibility criteria ahead of a downturn.

2 Operationalization roadmap

Phase	Action	Owner / dependency	Timeline

Phase 1	Deploy the uptake scoring pipeline as a batch job against the production loan book; validate scores against held-out 2023 outcomes	Data science + MLOps	Sprint 1–2
Phase 2	Implement the Markov risk-tier tracker as a scheduled job recomputing tier transitions every 60 days	Data engineering	Sprint 3–4
Phase 3	Stand up the optimization engine as a decision API (grant / hold / decline) integrated with the loan servicing system	Platform engineering	Sprint 5–6
Phase 4	Launch the portfolio risk dashboard and macro-trigger monitor for ongoing NPV and default-rate tracking	Risk & analytics	Sprint 6–7
Ongoing	Monthly NPV reporting cadence; quarterly strategy review incorporating updated macro indicators and refreshed model retraining	Lending strategy team	Quarterly

3 Summary scorecard

Metric	Value
Total customers analyzed	**30,000**
Eligible for increase	**25,068 (83.6%)**
Selected by optimization engine	**9,071**
Best model / ROC-AUC	**Random Forest / 0.5010**
Best strategy / mean NPV per customer	**Strategy B (Selective) / $77.13**
Total portfolio NPV (base case)	**$1,243,712**
Recession-scenario NPV impact	**−56.2%**

CLOSING NOTE

This framework is designed to be re-parameterized rather than rebuilt: every assumption documented in Section 2 (default rates, loss severity, capital ceiling, repayment mix) is an explicit variable in the implementation notebook. As production data on actual default rates, loss severities, and behavioral uptake drivers becomes available, those figures should directly replace the calibrated assumptions used throughout this analysis.